

5. Hardware Synchronization: Test-and-Set & Swap

Syllabus Coverage: Gateway Classes Slides 40–59 | Software Lock Flaw, Atomic TSL Primitives, Swap Instructions, Spinlock Evaluation.

5.1 The Software Lock Variable Flaw

Preemption between the `test ( while(lock==1) )` and `set ( lock=1 )` operations allows multiple processes to enter the Critical Section simultaneously, violating Mutual Exclusion.

5.2 Hardware Atomic Test_and_Set (TSL)

```
// Hardware Atomic Primitive
boolean Test_and_Set(boolean *target) {
    boolean r = *target;
    *target = true;
    return r;
}

// Process Synchronization Code
while (Test_and_Set(&lock)); // Spinlock busy waiting
/* CRITICAL SECTION */
lock = false; // Release lock
```

Hardware Synchronization: Test-and-Set (TSL) & Swap

Atomic Instruction Execution & Spinlock Mechanics

Why Simple Software Lock Fails (Flaw)

```
while (lock == 1); // 1. Check lock
// <--- PREEMPTION OCCURS HERE! --->
lock = 1; // 2. Set lock
/* CRITICAL SECTION */
```

The Flaw:

Agar lock 0 tha aur P1 ne test karke preemption le li, toh P2 bhi lock 0 dekh kar CS me chala jayega! Mutual Exclusion FAILS!

Hardware Atomic Test_and_Set (TSL)

```
boolean Test_and_Set(boolean *target) {
    boolean r = *target; // Read original
    *target = true; // Set to 1
    return r; // Return original
} // <--- EXECUTED IN 1 ATOMIC CLOCK CYCLE!
```

Usage in Code:

```
while (Test_and_Set(&lock)); /* CS */ lock = false;
```

Spinlock Architecture & Criteria Verification

What is a Spinlock?

• A lock where waiting thread spins continuously in while loop.

Good for: Short waiting time.

Bad for: Long wait (wastes CPU).

The Swap Instruction

```
void Swap(boolean *a, *b) {
    temp=*a; *a=*b; *b=temp;
}

key = true;
while(key) Swap(&lock, &key);
```

Criteria Evaluation

• Mutual Exclusion: ✓ YES

• Progress: ✓ YES

• Bounded Waiting: ✗ NO!

Random scheduling may cause a process to starve!

5.3 Criteria Satisfaction Summary

Criterion	Status	Technical Reason
Mutual Exclusion	✓ YES	Atomic memory bus lock prevents concurrent false returns.

Progress	✓ YES	No external or remainder process blocks CS entry.
Bounded Waiting	X NO	Random hardware spin contention cannot guarantee bounded turns (Starvation possible).